

PHYS 4319 Advanced Mechanics

Spring 2025

Instructor: Amir Shahmoradi, Joshua Osborne

Email: a.shahmoradi@uta.edu, joshua.osborne@mavs.uta.edu

Faculty Profile: www.cdslab.org

Lectures: Tuesday/Thursday 3:30-4:50 pm

Office Hours: Tuesday/Thursday 2:30-3:30 pm on Teams (or by appointment)

Course Description:

We will cover various topics in advanced mechanics, including oscillations (damped and driven), gravitation using differential calculus, Lagrangian and Hamiltonian dynamics, dynamics of systems of particles, and collision theory.

Where to find course material: 4319 Teams Channel

Textbook(s):

- 1) Morin, 2007, Introduction to Classical Mechanics.
- 2) Thornton, Classical Dynamics of Particles & Systems, 5th Edition.

Course Prerequisites:

- 1) PHYS 2311 (MATHEMATICAL METHODS OF PHYSICS),
- 2) PHYS 3321 (INTERMEDIATE ELECTRICITY AND MAGNETISM),
- 3) MATH 3319 (DIFFERENTIAL EQUATIONS & LINEAR ALGEBRA)
- 4) MATH 3318 (DIFFERENTIAL EQUATIONS)
- 5) or Permission of the Instructor.

Student Learning Outcomes:

In this course, you will develop a working knowledge of important concepts in advanced classical mechanics. You will learn how these concepts impact our understanding of the motion of single bodies, systems of bodies, and continuous systems under different conditions (e.g., subject to gravitational forces). By pulling in examples from a variety of areas of physics, you will come to appreciate their wide range of applications, from astrophysics to high-energy physics and many more. Upon mastering the material, you will have a deep appreciation for many of the core foundations of physics and know how to solve important problems related to these topics. The lectures, textbook readings, and homework problems are all designed to help you achieve these goals, and your success in all of these learning outcomes will be measured by your ability to answer in-class concept

questions and to apply the material covered in lecture and reading to solve homework and exam problems.

Grading:

- 1) Homework: 25%
- 2) Quizzes / Midterms: 40%
- 3) Final exam: 35%
- 4) Class participation: 5% extra credit (I am very good at remembering faces!)

Letter Grades:

Grand Score	Letter Grade
85 <= grade	A
75 <= grade < 85	B
65 <= grade < 75	C
55 <= grade < 65	D

Homework:

Starting the second week of the semester, there will be weekly homework assignments. Each homework will be assigned on Tuesday and is due next Tuesday at 3:30 p.m. before the next lecture begins.

Quizzes:

Periodic quizzes or exams will be held, and their dates, frequency, and structure will be determined based on class consensus.

Lectures:

Lectures will be pre-recorded and offered to you each Friday. You must watch the lectures, understand the material, optionally read the relevant sections of the textbook(s), and come to class next Tuesday with any questions you may have about the topic. We will quickly review the topic within the class and begin solving practice problems and homework questions together.

Course Expectations and Suggestions:

- 1) Attending and participating in lectures will greatly facilitate learning the material and problem-solving techniques.
- 2) I will post my lecture notes so you can review them later or on the rare occasion you need to miss a lecture. However, if attendance drops off significantly, I may stop posting these.
- 3) Besides lectures, you should spend 5-10 hours per week on course-related activities, including reading the textbook, completing assignments, and preparing for exams.

- 4) When you read the textbook, stop and think about the material, even taking notes as you go. I also recommend reading each chapter at least twice. The concepts will likely be a lot clearer on the second pass.
- 5) Many students find it very helpful to study with classmates, and I encourage you to do so. You are also encouraged to work together on homework problems. However, make sure you first try to answer the questions independently. You will learn the material better by engaging and even struggling with homework questions on your own first.
- 6) Please email me with questions, or even better, come to my office during office hours. If you are having difficulty with any of the course content or if any personal issues come up during the semester that are affecting your studies, please let me know.

University-wide policies:

Attendance: At The University of Texas at Arlington, taking attendance is not required, but attendance is a critical indicator of student success. Each faculty member is free to develop his or her own methods of evaluating students' academic performance, including establishing course-specific attendance policies. As the instructor of this section, I will be taking attendance periodically, and I strongly encourage all students to attend all classes. However, while UT Arlington does not require instructors to take attendance in their courses, the U.S. Department of Education requires that the University have a mechanism in place to mark when Federal Student Aid recipients "begin attendance in a course." UT Arlington instructors will report when students begin attendance in a course as part of the final grading process. Specifically, when assigning a student a grade of F, faculty report the last date a student attended their class based on evidence such as a test, participation in a class project or presentation, or an engagement online via Blackboard. This data is reported to the Department of Education for federal financial aid recipients.

Drop Policy:

Students may drop or swap (adding and dropping a class concurrently) classes through self-service in MyMav from the beginning to the late registration period. After the late registration period, students must see their academic advisor to drop a class or withdraw. Undeclared students must see an advisor in the University Advising Center. Drops can continue through a point two-thirds of the way through the term or session. The students are responsible for officially withdrawing if they do not plan to attend after registering. Students will not be automatically dropped for non-attendance. Repayment of certain types of financial aid administered through the University may be required due to dropping classes or withdrawing. For more information, contact the Office of Financial Aid and Scholarships (<http://wweb.uta.edu/aao/fao/>).

Disability Accommodations:

UT Arlington is on record as being committed to the spirit and letter of all federal equal opportunity legislation, including The Americans with Disabilities Act (ADA), The Americans with Disabilities Amendments Act (ADAAA), and Section 504 of the Rehabilitation Act. All instructors at UT Arlington are required by law to provide “reasonable accommodations” to students with disabilities so as not to discriminate based on disability. Students are responsible for providing the instructor with official notification in a letter certified by the Office for Students with Disabilities (OSD). Only those students who have officially documented a need for accommodation will have their request honored. Students experiencing a range of conditions (Physical, Learning, Chronic Health, Mental Health, and Sensory) that may cause diminished academic performance or other barriers to learning may seek services and/or accommodations by contacting:

The Office for Students with Disabilities, (OSD)

www.uta.edu/disability or calling 817-272-3364. Information regarding diagnostic criteria and policies for obtaining disability-based academic accommodations can be found at www.uta.edu/disability.

Counseling and Psychological Services, (CAPS)

www.uta.edu/caps/ or calling 817-272-3671 is also available to all students to help increase their understanding of personal issues, address mental and behavioral health problems, and make positive life changes.

Non-Discrimination Policy:

The University of Texas at Arlington does not discriminate on the basis of race, color, national origin, religion, age, gender, sexual orientation, disabilities, genetic information, and/or veteran status in its educational programs or activities. For more information, visit uta.edu/eos.

Title IX Policy:

The University of Texas at Arlington (“University”) is committed to maintaining a learning and working environment that is free from discrimination based on sex in accordance with Title IX of the Higher Education Amendments of 1972 (Title IX), which prohibits discrimination based on sex in educational programs or activities; Title VII of the Civil Rights Act of 1964 (Title VII), which prohibits sex discrimination in employment; and the Campus Sexual Violence Elimination Act (SaVE Act). Sexual misconduct is a form of sex discrimination and will not be tolerated. For information regarding Title IX, visit www.uta.edu/titleIX or contact Ms. Jean Hood, Vice President, and Title IX Coordinator, at (817) 272-7091 or jmhood@uta.edu.

Academic Integrity:

Students enrolled all UT Arlington courses are expected to adhere to the UT Arlington Honor Code:

“I pledge, on my honor, to uphold UT Arlington’s tradition of academic integrity, a tradition that values hard work and honest effort in the pursuit of academic excellence. I promise to submit only work that I personally create or contribute to group collaborations, appropriately referencing any work from other sources. I will follow the highest standards of integrity and uphold the spirit of the Honor Code.”

UT Arlington faculty members may employ the Honor Code in their courses by having students acknowledge the honor code as part of an examination or requiring students to incorporate the honor code into any work submitted. Per UT System Regents’ Rule 50101, §2.2, suspected violations of the university’s standards for academic integrity (including the Honor